BMC Medicine

RESEARCH

Open Access

The nature and frequency of food and beverage marketing on Kenyan national television: a mixed-method analysis of food advertisements, parent and children's perspectives

Milkah N. Wanjohi[1,2*], Caroline H. Karugu[3,4], Charles Agyemang[4,5], Michelle Holdsworth[6], Amos Laar[7], Kerstin Klipstein-Grobusch[2,8], Veronica Ojiambo[3,11,12], Sharon Mugo[3], Elizabeth W. Kimani-Murage[1], Stefanie Vandevijvere[9] and Gershim Asiki[3,10]

Abstract

Background Exposure of children and adolescents to unhealthy food through marketing and advertising on television (TV) is associated with increased consumption of unhealthy foods and subsequently, overweight/obesity and diet-related non-communicable diseases (NCDs). This study assessed the nature, frequency, and exposure of children to food and beverage advertisements on TV and the perspectives of children/adolescents and parents on food marketing on TV.

Methods Mixed methods, nationally representative study, guided by the International Network for Food and Obesity/NCDs Research, Monitoring and Action Support (INFORMAS) protocol for monitoring food promotion. This entailed simultaneous recording of the three most popular national TV channels in Kenya for eight randomly selected days, 18 h/day, over a 3-month period in 2021–2022. The NOVA classification was used to categorize food based on level of processing. Differences in advertisements by food groups, recording days (weekday/weekend) and seasons (holiday/non-holiday) were assessed. Focus group discussions ($n=14$) and in-depth interviews ($n=29$) were conducted with school children/adolescents (~ 9–18 years) and parents respectively in three counties, to explore their experiences and perspectives of food advertisements. Data were coded in NVIVO and analyzed thematically.

Results Of the 3700 advertisements recorded, about a third (36%, $n=1316$) comprised of food and beverages; 94.7% ($n=1213$) were ultra-processed foods (UPFs), with a mean rate of 2.8 ads/channel per hour, the majority (95.9%, $n=557$) of which were broadcast during peak hours. Some of the children interviewed vividly remembered some of the advertised food brands, mainly those related to UPFs. Most parents acknowledged that their children paid attention to food advertisements and sometimes requested and expressed preference for the advertised foods or brands. Parents generally considered food advertising to be safe, with no concerns about unhealthy food exposure to children.

*Correspondence:
Milkah N. Wanjohi
njerimilka230@gmail.com
Full list of author information is available at the end of the article

Conclusions Exposure of Kenyan children and adolescents to unhealthy foods through advertisements on national TV is higher than healthier food options. These advertisements enhance recognition and recall of the food brands, while parental concern about unhealthy food advertising is limited. Policies restricting advertising of unhealthy foods accompanied by nutrition education are urgently needed to limit unhealthy food exposure and consumption by children and adolescents, to address the increasing burden of overweight/obesity and NCDs.

Keywords Food environment, Marketing, Advertisements, Children, Adolescents, Kenya, Overweight/obesity

Background

Globally, overweight and obesity among children and adolescents have increased substantially, with an estimated 400 million children and adolescents aged 5–19 years living with overweight or obesity, a tenfold increase from 1990 [1].Current projections indicate a concerning trend: by 2030, 30% of children globally are expected to live with overweight or obesity. This increase is anticipated to be even more significant in low- and middle-income countries (LMICs), where the challenge of undernutrition persists [1, 2]. In Kenya, for example, the prevalence of overweight and obesity among adolescent girls has risen from 9 to 13% in the past 10 years. Simultaneously, a significant proportion (43%) of adolescent boys in the country are underweight [3]. Child overweight and obesity have adverse and intergenerational consequences, including increased risks of metabolic, cardiovascular, mental, psychological, neurologic, and digestive diseases/disorders [1, 4, 5]. Consumption of unhealthy diets, including ultra-processed foods (UPFs), and energy-dense and nutrient-poor foods is one of the leading drivers of overweight and obesity [1, 6–8]. Obesogenic environments that include marketing and advertising of unhealthy foods fuel the exposure to and consumption of unhealthy diets by children and adolescents [4, 6, 9]. A global benchmark of children's exposure to unhealthy foods through TV advertisements revealed a fourfold higher prevalence of unhealthy food advertisements compared to those for healthier food options [10], most frequently during children's peak viewing times [10]. There is evidence that food and beverage marketing influences children and adolescents' food choices, purchase requests, and increased intake of unhealthy foods [11, 12], thus exposing them to increased risk of NCDs and related poor health outcomes in the long term [9].

TV is one of the leading media sources of food advertisements for children/adolescents [13, 14]. In Kenya, TV is the most dominant medium [15], owned by 70% of rural and 80% of urban households. Over 80% of those who own a TV watch it daily [15], and about 46% of the content accessed includes advertisements [16]. In addition, mobile phones and computer devices are used to access TV channels by 23% of Kenyans in both rural and urban areas [16]. The majority of school-going children and adolescents in Kenya eat their meals while watching TV, often unsupervised [17].

In 2010, the World Health Assembly (WHA) Resolution 63.14 was adopted to guide countries in developing policies to reduce the exposure of children to the marketing of unhealthy foods and beverages that are high in fats, sugars, and salt, and to decrease the negative impacts of such marketing on children's health [18]. The World Health Organization (WHO) Commission on Ending Childhood Obesity (ECHO) recommends reducing children's exposure to the marketing of unhealthy foods on various media platforms [19]. However, the implementation of these recommendations has been slow, particularly in sub-Saharan Africa [14, 20]. In Kenya, while regulations exist for the advertisement of alcohol and breastmilk substitutes, there are currently no policies specifically restricting the marketing of unhealthy foods to children and adolescents on television. This gap may be due to limited research in sub-Saharan Africa on the extent and impact of such advertising.

To address this, our study assessed the nature and frequency of unhealthy food advertisements on Kenyan national TV channels. We also explored the perspectives of children and parents regarding these advertisements across three major regions in Kenya. The findings from this research aim to provide evidence that can inform the development of policies to limit children's exposure to TV advertisements for unhealthy foods.

Methods

Study design

This cross-sectional mixed-method study was conducted between December 2021 and February 2022. Quantitative methods involved recording popular national TV programs to identify the type and frequency of foods advertised on TV. The methodology was adopted from the International Network for Food and Obesity/NCDs Research, Monitoring and Action Support (INFORMAS) protocol for monitoring the nature and extent of the promotion of unhealthy foods and beverages on TV [21]. The qualitative methods involved focus group discussions (FGDs) and in-depth interviews (IDIs) with school-going children/adolescents (9 to 18 years) and parents respectively in selected schools from three counties in Kenya to explore their perspectives and experiences of food advertisements on TV.

Population and sample

Quantitative study (TV monitoring)

A three-stage sampling approach was used to select the TV stations and the days for recording programs.

Identification of TV stations for recording Purposive sampling was used to select the three most popular national digital TV channels in Kenya using the Communication Authority of Kenya's (CAK) and Geopoll's reports, which provided annual viewership ratings for TV and other media [22, 23] in Kenya. We used their reports to identify the three TV stations with the highest viewership of both adults and children, free to air, and with national coverage. Pay-to-air and TV with only regional coverage were excluded to minimize bias, as they were not available or accessible to all Kenyan citizens. Using these selection criteria, Citizen TV, NTV, and KTN were included in the study, as they had the highest rating and were most popular with children and adults in 2019–2020 [22, 23].

Selection of recording months, days, hours Based on INFORMAS guidelines for monitoring the exposure of unhealthy foods to children through TV advertisements [21], a three-stage sampling procedure was applied. First, the recording months were selected purposively, where we selected three consecutive months for recording.

To capture potential variations in food marketing and advertising, a school holiday (December 2021) and a school term (January and February 2022) were selected.

Secondly, stratified sampling was applied to select the recording days, with weekdays and weekend days comprising the two strata. In the third stage of sampling, four weekdays and four weekends were randomly selected spanning the 3-month period (December 2021–February 2022). National holidays, special national events, and public holidays were excluded before sampling to ensure a typical broadcasting setting in the country. Finally, for each day selected, 18 consecutive hours of recording were selected from 06:00 to 24:00.

TV data recording The peak viewing times for children and adolescents was estimated as between 5 and 9 pm based on the Communication Authority of Kenya (CAK) reports [16]. Three project team members recorded the three TV stations concurrently, on each of the selected month/day/hour using a digital TV with an inbuilt audio/visual recorder (TCL 32 screen inch). The recorded data was saved on a laptop and backed up on a portable hard drive after completion of the recording to avoid any data loss.

TV data extraction and coding

A data extraction form adopted from the INFORMAS guidelines [21] was programmed on the Survey for Computer-Assisted Technology and Operations (SurveyCTO) and used to identify and code information on all food and non-food advertisements including commercials and product placements appearing in the respective channels. The information coded includes the TV channel name, date, and day/date of recording, time slot when the advertisement appeared, program category and names, type of advertisements (food or non-food, brand names without food items), the specific foods advertised, brand names of the products, and any marketing/persuasive techniques used. The marketing techniques/persuasive techniques included promotional characters and premium offers. The promotional characters included cartoon or company-owned characters, the use of children/adolescents and parents in posts, use of celebrities in posts, while examples of premium offers were price offers/discounts, and "buy one get one free". The various program category names considered included news, sport (a specific program or sport), soap opera, series (not specifically for children), movie (not specifically for children), documentary, reality show, talk show, miscellaneous entertainment, e.g., variety, children (e.g., cartoon, movies, series), music or music video, religious, and health-related programs [21]. Each channel was coded by two research assistants and reviewed by the project team for reliability and consistency checks. Any discrepancies in the coded data were discussed and addressed by project members. Inter-rater reliability checks for key variables including categorization of advertised products as either food or non-food and the number of food items advertised were conducted to compare the agreement rate between two coders for each TV channel.

TV data analysis

Advertised foods were categorized using the NOVA food classification, which categorizes foods based on the level and nature of industrial processing into four groups: (i) unprocessed or minimally processed foods, (ii) processed culinary ingredients, (iii) processed foods, and (iv) UPFs. UPFs are industrially formulated edible substances derived from natural foods or synthesized from other organic compounds that are industrially processed and are rarely used for normal kitchen cooking, such as artificial sweeteners, flavors, food colors, hydrolyzed proteins, emulsifiers, thickeners, carbonating, gelling, bulking, foaming, and glazing agents. They represent the most unhealthy food group in the NOVA classification [24]. We excluded alcohol from the NOVA classification and treated it as a separate group since the focus was on non-alcoholic food and beverage products.

The mean (SD) of food advertisement rates per channel-hour was computed across all 144 h for each channel. To account for skewness in ad placement, median and interquartile range (IQR) were also calculated, limited to hours with at least one food advertisement. Frequencies (%) were used to describe the overall advertisement categories and the food groups. A Chi-Square Goodness-of-Fit test was conducted to assess whether total advertisements and food and beverage advertisements were evenly distributed across the three television channels. These were analyzed for the TV channel and peak viewing time for children. The Kruskal Wallis test was used to compare the rates of the four NOVA food group advertisements by TV channels, peak times, recording day (weekday/weekend), and season (holiday/non-holiday). Stata Version 15 [25] was used for data analysis, and the statistically significant level was set at $\alpha = 0.05$.

Qualitative study

The qualitative study focused on exploring the perspectives and experiences of children, adolescents, and parents on TV food advertisements. Participants were selected from three major counties in Kenya: Nairobi, Mombasa, and Baringo; representing an urban major/capital city, an urban secondary city/major tourist destination, and a rural/agricultural county, respectively.

Selection of study schools and participants Multi-stage sampling was used to select both low and high socioeconomic status (SES) sub-counties in each county based on the household budget survey report [26], after which three schools were randomly selected in each sub-county. School selection ensured representation of both primary and secondary schools, private and public schools, to obtain a wide range of experiences and perspectives from different age groups and settings. In total, 29 schools were selected: 8 schools in Nairobi, 15 in Mombasa, and 6 in Baringo.

In each selected school, convenience sampling was used to select students to participate in the FDGs. All students in secondary schools (year 9 to 12) were eligible to participate, but in primary schools, only students in grades (years) 4 to 8 were eligible to participate, as the students in these grades have an average age of 9 to 14 years and were perceived by the researchers as able to express themselves comfortably in group discussions compared to younger students in lower grades, based on previous experience. In addition, at least two parents (male/female) were sampled from each study school to participate in in-depth interviews. School heads and teachers assisted in the identification of eligible students and parents.

Focus groups with children/adolescents and interviews with parents FGDs were conducted with 6–8 students per group, lasted about 45 to 60 min, and were conducted within schools. This is the maximum number of students per group that could be allowed by the national COVID-19 prevention guidelines, which restricted the number of people in any group meeting not to exceed 8 and meeting duration to less than 1 h.

The discussions were facilitated using an open-ended interview guide. IDIs were conducted with parents from the selected schools using an open-ended interview guide. The focus group and interview guides were based on the research questions and adaptations from existing literature [27, 28]. They were further refined after the pilot exercise to fit the study contexts (see Additional file 1). Trained qualitative research assistants facilitated the group discussions and interviews, in English or Swahili, based on the participant's preference.

Qualitative data management and analysis IDIs and FGDs were recorded using a digital audio recorder and later translated and transcribed verbatim into Microsoft Word. The transcripts were coded in NVIVO software. Thematic analysis with both deductive and inductive approaches was applied during data coding and analysis. The deductive approach was based mainly on the research questions, while the inductive approach involved the identification of new themes generated from the data beyond the main questions in the interview guide. During the data analysis and synthesis process, prominent and commonly occurring concepts and ideas were identified and organized into themes [29]. A framework matrix of the coded data was developed to allow for comparison across the counties, sub-counties, and school types.

Results

Overall advertisement characteristics

In total, 3700 advertisements were recorded for a total of 432 h; 35.6% ($n = 1316$) comprised food/beverage advertisements (Table 1). The frequency (percentage (%)) and the mean rate of advertisements were determined (Table 1). A Chi-square Goodness-of-Fit showed significant variation in the distribution of total advertisements, χ^2 (2, $N = 3700$) = 153.50, $p < 0.0001$, as well as in the distribution of food and beverage advertisements, χ^2 (2, $N = 1316$) = 25.72, $p < 0.001$. These results indicate that

Table 1 Overall advertisement frequencies by TV channels

Channels	Total (all) Advertisements n (%)	*P*-value	Total food and beverages advertisements n (%)	*P*-value	Mean rate of advertisements (ads per channel-hour) (SD)	
	(N=3700)		(N=1316)		All Advertisements	Food and beverage advertisements
NTV	1352 (36.5%)	0.0001	498 (37.8%)	< 0.001	9.4 (28.5)	3.4 (11.1)
Citizen TV	1464 (39.6%)		464 (35.2%)		10.2 (32.7)	3.4 (11.3)
KTN	884 (23.9%)		354 (26.9%)		6.1 (18.3)	2.5 (7.8)

both overall advertising volume and food and beverage-specific advertisements were disproportionately represented across NTV, Citizen TV, and KTN. The overall mean rate (of advertisements per channel per hour) was 8.6, while for food/beverage advertisements, this was 3.0. NTV and Citizen TV had higher and similar patterns of advertising, and KTN had the lowest frequency of advertising per channel hour (Table 1).

Participant characteristics (qualitative study)

In total, 14 FGDs were conducted with school children and adolescents (9 to 18 years), while 29 IDIs were conducted with parents in the three counties (Table 2).

Type of foods advertised

Across all broadcasting channels, nearly all (95%, $n=1213$) of the food and beverage products advertised were classified as UPFs. The overall mean rate of advertisements of UPFs per channel per hour was 2.8. The ratio of UPFs to unprocessed/minimally processed foods was 3:1.

Minimally processed foods were less frequently advertised; staples including maize, wheat, and rice were the most common foods (3.6%) advertised in this category. Among the UPFs, sugar-sweetened drinks (58.4%, $n=748$), chocolate, sweets, and candy (15.5%, $n=199$), and fast food (12.4%, $n=159$) were the most frequently advertised foods and beverages (Table 3). Food classification using NOVA classification is shown in Table 3.

Children's perspectives on type of foods advertised

Most children acknowledged paying attention to food advertisements on TV. The majority remembered and described in detail the TV food advertisements they liked and remembered, including the food items, brand names, characters, songs, activities, information, and offers appearing in the advertisements.

From the discussions on food advertisements "watched recently," and "liked/remembered most" by the children, UPFs were the commonly mentioned food items and brands. Examples include sugar-sweetened beverages, such as various brands and flavors of soda (Coca-Cola, Fanta, and Sprite), energy drinks (lucozade and red-bull), and commercial fruit juices. Other foods included yoghurt, breakfast cereals (weetabix, cornflakes) bread, buns, bread rolls, noodles (Indomil) condiments (royco), sausages, Frenchfries (chips), cakes, spreads such as margarine (blueband), jam, mayonnaise, and chewing gum.

> *"I like Festive bread adverts, about the quantity, price and even the color of the bread. Soda, they advertise that if you drink it, you get a lot of energy than usual...On the same advert, they advertise other snacks that someone can eat with soda, some buns or some cakes". (FGD mixed group, MK public primary school, Baringo).*

Non-UPFs were also mentioned, but less often; they included staples such as wheat, corn, and porridge flour; rice; beverages such as milk, bottled drinking water, and cooking oil. In a few cases, alcoholic drinks were also mentioned as part of the advertisements watched on TV.

> *" I have seen Fresh Fry cooking oil on television. Most of the time it is aired on Citizen TV, and I like it because it has no cholesterol, ...I have seen this EXE*

Table 2 Participant characteristics

Interview type	Number of interviews			
	Nairobi	Mombasa	Baringo	Total
Focus group discussions (FGDs) with younger adolescents (9–14 years)	4	3	4	11
FGDs with older adolescents (15–18 years)	1	2	0	3
In-depth interviews with parents	6	15	8	29

Table 3 Overall food groups/categories of food advertisements by NOVA categories

Food groups/categories (*N* = 1309)	Frequency % of advertising	
UPFs (*N* = 1213)		
Sugar-sweetened drinks (soft drinks)	748	58.4%
Chocolate and candy	199	15.5%
Fast food (french fries, burgers, etc.)	159	12.4%
Instant noodles, rice, and products with, added fat, sugar, and salt	38	3.0%
Milk and yogurts and their alternatives with added sugars	31	2.4%
Meat and meat alternatives (processed)	14	1.1%
Recipe additions (e.g., soup cubes)	10	0.8%
Savory sauces and spreads	6	0.5%
Fruit juice/drinks < 98% fruit	3	0.2%
Savory snack foods with added salt or fat	2	0.2%
Sweetbreads, cakes, muffins, sweet buns and sweet snack foods	2	0.2%
Tea and coffee products with sweeteners	1	0.1%
Unprocessed/minimally processed foods(*N* = 55)		0.0%
Wheat, rice, and rice products without added fat, sugar, and salt,	46	3.6%
Natural Milk and yogurts	1	0.1%
Meat and meat alternatives—include meat, poultry, fish, legumes, tofu, eggs, and raw unsalted nuts (unprocessed)	4	0.3%
Tea and coffee products with no additives	4	0.4%
Processed culinary ingredients (*N* = 13)		
Recipe additions include dried herbs, and seasonings -Natural form	12	0.9%
Cooking oils	1	0.1%
Alcohol	35	2.7%

flour for porridge, it shows a family from father to a child taking porridge made from that flour...I liked the one on Nutri Porridge, it was inspiring". (FGD mixed group, RC public primary school, Nairobi).

The advertisements discussed were mainly for single food items but also food combinations and recipes. The advertisement content included information on product varieties, such as flavors and colors, quantities, sizes, and prices. Participants also reported on "non-food" adverts that included food content, for example, food delivery company advertisements that included fast foods (e.g., pizza, french fries) and restaurants that were associated with fast foods (e.g., *KFC*).

"The advertisement of Glovo (food delivery app), when you have the App of Glovo you can call them and order anything that you want e.g., pizza, chicken, then they will bring to you anytime" (FGD with girls, SSS public secondary school, Mombasa).

"I have seen the one on KFC; it makes one desire to buy their products because they have made a lot of effort to produce it... KFC even has its own products like ice cream, their Kentucky chicken, chips and all that I feel it is something you can eat for the moment because they advertise very good things. They show you how they are making it but of course, they cannot show you the recipes. They will show you how one eats and feels and that makes you start salivating" (FGD mixed group, RC public primary school, Nairobi).

The concept of "junk" food emerged from the discussions with a few participants (children and adolescents) from Nairobi and Baringo, indicating that the foods advertised were so-called "junk" and unhealthy, with a description of junk foods as those high in sugar or fat.

"What I don't like about their products is that a lot of it is junk. it is unhealthy ... for example the chicken it has a lot of oil and when you eat it you will have consumed a lot of fat... Lucozade, it says that you can get energy when you drink it... but you might find that it is 'junk' food, you might find that it is not healthy food" (FGD mixed group, RC public primary school, Nairobi).

"[Advert I remember most] is about junk foods, pizza, burger, hotdog and many others... Because it is not a healthy food" (FGD with girls, NP public primary school, Baringo)

Persuasive techniques

Our analysis of 1,281 non-alcoholic food and beverage advertisements on television revealed several common persuasive techniques. The most frequent were historical and cultural references, such as Christmas and Ramadan, which appeared in 277 (21.6%) of the advertisements. We also noted the use of movie tie-ins in 76 (5.9%) ads, sports-related themes in 42 (3.3%), and licensed characters in 40 (3.1%). Interestingly, our findings indicate that these persuasive strategies were mainly used to market ultra-processed food (UPF) products. Specifically, 270 (97.5%) of the ads using historical events, 71 (93.4%) of the movie tie-in ads, 27 (64.3%) of the sports-themed ads, and all 40 (100%) of the ads featuring licensed characters were for UPF products.

As the children and adolescents described the food advertisements that they remembered or liked most, and the reasons for that, several persuasive marketing and advertisement techniques were noted and organized into seven categories as summarized in Table 4. They included the following: (i) children/family themed advertisement "you see a family very happy taking their meal accompanied with Coca Cola drink"; (ii) promotional cartoons/characters "Red Bull, which makes him have wings to fly"; (iii) fun/happiness themed advertisements "There are children taking the yogurt and singing"; (iv) local celebrities "*… [*Local actress] the character in the advert happens to be my favorite artist"; (v) product appealing qualities "I like the advert because it is soft, the bread is also soft to eat"; (vi) product health and nutrition claims "if you use that Blue Band you can get the energy to play, work and even you can have energy for swimming"; and (vii) price offers, premiums and free gifts "Blue Band with the scratch, we used to rush for that scratch card so that you can be able to earn something". In some instances, a combination of techniques was identified (Table 4).

Advertisement time/period and trends

The analysis of advertising frequency revealed some interesting patterns. We observed a slightly higher occurrence of UPF advertisements during peak viewing times (95.9%, $n=557$) compared to non-peak times (93.3%, $n=656$), and also more frequently on weekdays (96.2%, $n=583$) versus weekend days (93.3%, $n=623$).

Conversely, advertisements for non-UPFs appeared less often during peak viewing hours (3.0%, $n=18$) compared to non-peak hours (5.3%, $n=37$), and were more prevalent on weekends (6.0%) compared to weekdays (2.5%). Furthermore, UPFs were advertised at a high rate during non-school holiday periods (95.7%, $n=751$), as detailed in Table 5.

The mean rate of UPF advertisements per channel hour was 1.3 during peak viewing times, 1.5 on weekends, and 1.1 during the holiday season.

Regarding the 18-h advertisement trend, the frequency of UPF advertising was higher than other food groups throughout the day, with the highest level of advertisements from 06:00 am to 21:00 pm, which corresponds to the children's peak viewing times in Kenya (Fig. 1).

From the qualitative discussions, most children and parents reported seeing the adverts in the evening, and less in the morning or during the day. The participants recalled seeing more adverts appear in the midst of popular local TV programs and just before, during, or after news broadcasts. In a few cases, participants reported having watched the adverts on non-local TV stations or programs, especially those in Nairobi and Mombasa, and less so for Baringo. In addition, participants highlighted that some of the foods were frequently advertised throughout the day and hence easily remembered.

> *"I have seen Yego biscuits, and they said it was sweet, it is on Citizen TV and during news break, I love sugary things" (FGD mixed group, RC public primary school, Nairobi).*
> *"Fanta, just before the program on the Brave and Beautiful program starts …Coca Cola, after a break or after news… [Most memorable advert] Coca-cola, because they advertise it every day… The one on Lucozade, because it is always being advertised every day…. Weetabix, it is always advertised every day" (FGD with boys, TP public primary school, Baringo).*
> *"I remember the one on Dola flour… Because it is all over…Coke, it is always being advertised all the time" (IDI student, MPP public primary school, Mombasa).*

Parents' views on TV advertisements to children

Some parents acknowledged that their children paid attention and liked the food advertisements on TV, sometimes singing along and imitating the songs/jingles in the advertisements. Parents also recounted that sometimes their children developed an interest and greater preference for the advertised foods compared to the homemade meals. In some cases, children requested, nagged about, or demanded the food items and brands watched in the advertisements, some of which parents gave in to.

> *"My children feel very good and that is why they have crammed the adverts and sing along, meaning that they are happy with the adverts. Yes, that is what makes them enjoy watching TV because they imitate what they see. For example the advert on flour, Ajab flour, when it is on air, they have crammed all those words so that they sing and dance along with the*

Table 4 Persuasive advertisement strategies identified from the discussions with children and adolescents

Children/family themed content/advertisement	"(Chocolate), it shows the secret of good upbringing. A mother goes to work leaving a child behind, then the grandmother comes and is given a smaller piece of chocolate. The grandchild takes the bigger piece and hides in the grandma's bag where she finds it later… I love the Coca Cola advert, it looks nice when the woman in the advertisement brings together her family for a meal and together enjoy the Coke drink as a family'joy moments', you see a family very happy taking their meal accompanied with Coca Cola drink. In the advert Ilara milk they say that you'll have energy and strength to think during boredom, when you take that milk, you kind of revive your mind. And again, in this advertisement, what is more fun, you see an old man, his wife and two children going to school and they cannot go without taking the milk first before heading to school. After taking the milk, the father is very happy playing with his children. I like that one part very much…" (FGD boys&girls, KH, public secondary school, Low SES sub-county, Mombasa) "There is one on Blue Band which is good for students. You see a student who has taken Blue Band for breakfast, he is able to do constructive things, a footballer grows strong, that is what I have seen about it" (FGD boys&girls, RC public primary school, Low SES sub-county, Nairobi)
Fun and happiness themed advertisements	**"**Blueband, there was a time, the twin boys took that in the morning when they were going to school, they were participating in the athletics competitions, one fell and the other one went and held his brother, so, I just like the way they helped each other. Yoghurt, I love watching the advertisement on Ilara Yoghurt in particular. What I like about that advertisement is that there are children taking the yoghurt and singing (FGD boys, BP public primary school, Low SES sub-county, Mombasa) "The advert I have seen on television is Kelloggs. Kelloggs resembles Weetabix but this one is made from corn, there is a programme on jogging sponsored by that Kelloggs, I love it so much, It [advert] says how strong it is, makes you strong and you grow mentally, physical and they can have energy. It shows how the child play after eating that Kelloggs… I feel good and wish every parent should buy it because it is strong but then it is expensive, low-class people cannot afford it. (FGD mixed group, RC public primary school, Low SES subcounty, Nairobi) "Sprite. I just love seeing Sprite and when I see it, I feel like drinking it… Because when they are advertising it, there is a character of a boy dancing" (FGD boys, KR public primary, Low SES subcounty, Baringo)
Celebrities	"There are two characters acting on this advertisement. The first one is Eric Omondi (local comedian] while there is a Swahili woman. So, they are pulling this kneaded flour like a towel… The food advertisement that I love most is the [one] of Indomie with Maria [Local actress], Supa Mojo Indomie with Maria… I like the Indomie advert because Maria the character in the advert happens to be my favorite artist" (FGD boys&girls, KH public secondary school, low SES sub county, Mombasa) "The advertisement that I remember most is the one on Blue Band which they were using teacher Wanjiku (local comedian) though it is no longer being aired" (FGD boys and girls, RC public primary school, Low SES sub-county, Nairobi)
Cartoons and characters	"Red bull, It shows that there are two antelopes and there is a lion coming, so one tells the other let me go fast, but the other one says no I want to be faster than you so that it can be eaten by a lion. So, I feel it's something more reliable, good, though I haven't tasted it" (FGD boys, BP public primary school, Low SES sub-county, Nairobi) "When they advertise for that product (milk) they will bring pictures of a cow, a normal baby and another calf with features of a normal baby dancing there, so they advertise how sweet that milk is, the child will tell you please buy for us that milk. You buy so that they can feel good like that small cow that appears in the advert" (IDI with parent, HM public primary school, Low SES sub-county, Nairobi) "I like the Dola flour has amusing cartoons and in the Blue Band advert" (IDI with student, MPP public primary school, Low SES sub-county, Mombasa) "Red Bull which makes him have wings to fly. …." (FGD mixed gender, KH public secondary school, low SES sub county, Mombasa) "There are those other things that he sees like those cartoons dancing. He really loves seeing them and he will buy the juice with those cartoons on it… Even when I cook that Ugali I have in the house he will not eat the way he is supposed to. If I sacrifice and cook what he has seen on TV; like the spaghetti, he will eat it, but he will not gain anything from it (less nutritious)" (IDI with parent, AM private primary school, Nairobi)

Table 4 (continued)

Product emotional appeal **Taste, flavor, color, convenience**	"Fanta, the colour and how fresh it is. I love it because when you take it, you become strong and it last for long, it doesn't expire fast and it is fresh" (FGD boys, TP public primary school, low SES sub-county, Baringo) "Bread, I like the advert because it is soft, the bread is also soft to eat" (FGD boys, BP public primary school, Low SES sub-county, Nairobi) "Chewing gum There are many different types, being advertised, we have the blue, green and yellow ones and each one has its own flavour…. I like the blue colour, it has peppermint flavour which I like very much" (FGD boys&girls, KH public secondary school, low SES sub county, Mombasa) "Like the bread, they advertise the bread called Festive …and the good thing they advertise about it is it has a closable seal like you can open then when you have finished using the bread you can close again" (FGD boys&girls, MKP public primary school, Low SES sub-county, Baringo)
Health and nutritional claims	"They advertise that if you use that Blue Band, you can get energy to play, work and even you can have energy for swimming. They advertise bread roll which has a chocolate inside it… they advertise for example if you were a herdsman, if you eat it, you will be able to take care of your cows very well…. soda. They advertise that if you drink it, you get a lot more energy than usual" (FGD, mixed gender, MKP public primary school, Low SES sub-county, Baringo) "The one I have seen is the one on Coca Cola…. Not everyone takes sugary staff and now at least they have brought Coca Cola that has not sugar and is good for even those people with diabetes… I have remembered the advert on Nescafe, it is some kind of coffee, whereby you just boil some water, put in the coffee, add sugar and then stir and take. That Nescafe is healthy, it is healthy, if you use it, you get energy. And you mind becomes focused, those coffee berries are natural and once in your body they provide energy and the strength and keeps your mind active" (FGD mixed group, RC public primary school, Low SES sub-county, Nairobi) "You see a child in a race with friends. When this child falls down, the mother gives him bread spread with Blue Band and the child get enough strength to run faster and wins the race… Fanta when they are advertising they show some youths seated there bored and not practicing but the moment they take that Fanta the get the moral (energy) to do their practices/games" (FGD mixed gender, KH public secondary school, low SES sub county, Mombasa)
Price offers, premiums and free gifts	"They say that when you buy Dola flour, when you open the packet you will find a number, which can give you a chance to win something" (IDI with student, MPP public primary school, Low SES sub-county, Mombasa) "Sometimes I can just sit in our home or even at school and just remember that there is a bucket of blue band which you can use…some come with a form, you fill it and then you can win prizes" (FGD boys&girls, MKP public primary school, Low SES sub-county, Baringo) "Blue Band with the scratch, we used to rush for that scratch card so that you can be able to earn something so that meant Blue Band was always at home, that is why I love it very much" (FGD mixed group, RC public primary school, Low SES sub-county, Nairobi)

advert…So when they see the Ajab flour, even if their mother is there, they will disturb her; 'mum see, Ajab is good flour, see that', for drinks, for example Pepsi or Coca Cola, when they are being aired, they also want to do the same as the people in the advert are doing. Like Coca Cola, the way it is advertised, they want to imitate those actions they are seeing on TV" (IDI with parent, JG public secondary school, Mombasa).

"They like seeing how the person drinks that Coke and after finishing the body vibrates…The younger one will just stress you because he will want you to buy that drink for him" (IDI, parent, GM private primary school, Mombasa).

"At times an advert may appear, and the children will tell you 'Mum why you don't buy that'. Let us go and buy that. For example, it can be an advert on pizza, chicken, or soda. Like soda when there is a sale, they will tell you to buy 2 and get one free, such like thing. Especially soda, the children will tell you we haven't taken it for a while and they have said if you buy 2 you get 2 free" (IDI parent, SC private primary school, Nairobi).

Parental monitoring and restrictions of food content on TV

From the narratives, some parents applied restrictions on TV stations and programs that their children could watch mainly due to "bad language," "bad behavior," "nudity," "gangster language," "violence," or "adult content." The restrictions were mainly applied to protect children from being influenced into "bad behavior."

Most parents reported having no concerns and hence applied no restrictions on food or nutrition-related content and advertisements as they found such content safe for their children, except those involving alcohol and cigarettes. Some parents perceived that the information in the advertisement was accurate and well-verified before it was approved for airing on TV.

Table 5 Distribution of food and beverage advertisements classified by NOVA food categories in peak vs non-peak time and day of the week

	Overall	**Peak viewing time**	**Non-peak viewing time**		**Weekend**	**Weekdays**		**Holiday season**	**Non-holiday season**	
	(*N* = 1281)	**(*N* = 581)**	**(*N* = 700)**		**(*N* = 668)**	**(*N* = 606)**		***N* = 496**	***N* = 785**	
NOVA food categories	***n* (%)**	***n* (%)**	***n* (%)**	***P*-value**	***n* (%)**	***n* (%)**	***P*-value**	*n* (%)	*n* (%)	***P*-value**
Ultra-processed foods	1213(94.7%)	557(95.9%)	656 (93.7%)	**0.0001**	623(93.3%)	583(96.2%)	**0.0001**	462 (93.2%)	751(95.7%)	**0.0001**
Unprocessed/minimally processed foods	55 (4.3%)	18 (3.1%)	37 (5.3%)	**0.0001**	40 (6.0%)	15 (2.5%)	**0.0001**	21 (4.2%)	34(4.3%)	**0.0001**
Processed culinary foods	13 (1.0%)	6 (1.0%)	7 (1.0%)	**0.0001**	5 (0.8%)	8 (1.3%)	**0.0001**	13(2.6%)	-	

Notes: The *P*-value is from Kruskal–Wallis (KW) tests. *P*-value: from KW test between NOVA food categories and the peak viewing times, weekday/weekend days, holiday and non-holiday seasons, %-column percentages showing the food categories in different day categorization

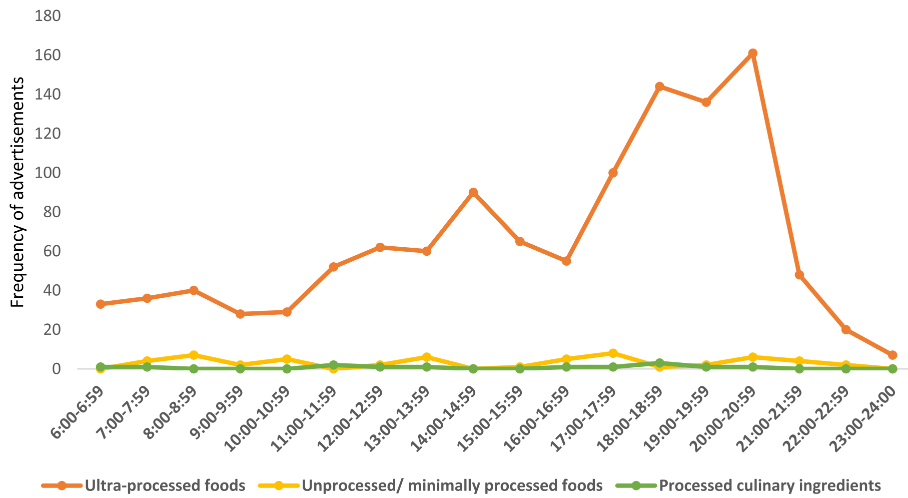

Fig. 1 Eighteen-hour trends of food and beverage advertisements by NOVA categories

> *"During the news bulletin, I do not restrict them, when there is a program that the child should not watch I don't allow them. Especially that one which is on relationships" (IDI parent, MP public primary school, Baringo)*

> *"School going children are listening to a family planning advertisement or an advert on alcoholic drinks, those ones, I usually advise them to change the channel because it will mislead… including those ones of cigarettes, alcohol" (IDI with parent, GP private primary school, Nairobi)*

> *"Me, always, I feel anything being advertised through the media is right. They cannot go for commercial advertisement before satisfying that these things are ok. To me I feel they are ok and that is why they are being aired, it is something good for our health, they should promote something that is good for our health and our children" (IDI parent, ST public primary school, Mombasa).*

Advertisements with age restrictions and warning information

Discussions on warning advertisements, mainly on alcoholic drinks, emerged from participants in Mombasa and Nairobi with children/adolescents and parents who appreciated the warning information and age restrictions provided in alcoholic beverages' advertisements. Some parents highlighted that the warning information provided an opportunity for engaging their children on the foods and content that were permissible (or not) for children and adolescents.

> *"The alcohol advertisement, it is indicated that a student is not allowed to take alcohol or whoever takes alcohol is supposed to drink responsibly. Because you can cause accidents on the road due to drunkenness" (FGD mixed groups, KH public secondary school, Mombasa).*

> *"The good thing about that commercial advertisement is it is quite exciting for him. For the older child, let us say he sees something like Tusker being advertised and it states very clearly not recommended for persons under eighteen years, he will not come and ask you about it or even find him using it because it is indicated there clearly, and he knows he is under eighteen years. But when he sees this soda being advertised, he knows that this one has no age restrictions. So, you see it really helps a lot" (IDI parent, AM private primary school, Nairobi).*

> *"It's the Kenya Breweries which advertised their beverages like Tusker, Guinness, and other spirits like Kibao. They talk of the age at which you can take alcohol, they also advise people to drink it responsibly and they also warn people of its health hazards, age limit is eighteen years and above" (IDI parent, SV private primary school, Mombasa).*

Discussion

This study aimed to investigate the characteristics and prevalence of food marketing aimed at children and adolescents on television. It also explored the viewpoints and experiences of both children and parents concerning food and beverage advertisements. Our findings indicated a significant presence of unhealthy food (UPFs) advertisements, averaging about three such advertisements per hour. Notably, unhealthy foods were advertised three times more often than healthier alternatives. We also observed a rise in the frequency of unhealthy food advertising as the day progressed, peaking during children's prime viewing times. The most frequently advertised categories were sugar-sweetened beverages, chocolates/sweets, and fast foods.

Conversations with children and adolescents showed that they notice, recognize, and remember the foods advertised on TV. In contrast, parents generally perceived this content as safe, with few restrictions or expressions of concern. Food and beverage advertisements employed various persuasive techniques that may effectively capture children's attention.

These results underscore the extent to which children in Kenya are exposed to unhealthy food marketing on national television. This exposure has the potential to shape their dietary behaviors, including their ability to recognize and recall unhealthy food brands, particularly given the limited monitoring and concern regarding these advertisements among parents.

The findings from our study regarding the high prevalence of unhealthy advertisements compared to healthier options align with findings from a global review which highlights the significant rates of unhealthy food advertising targeting children on television [10]. For example, a study in South Africa revealed that about 60% of the food advertisements were classified as unhealthy and were predominantly aired during peak viewing times [30]. Similarly, 84% of food advertisements in Italy were unhealthy [31], and in Argentina, two-thirds of foods advertised on TV were unhealthy [32].

Consistent with our findings, sugar-sweetened beverages were frequently advertised in the USA, Russia, and South Africa [30, 33, 34]. Furthermore, a review of commercial advertisements in Europe indicated that fast foods and savory snacks were commonly promoted [35] while sweets and confectionery were commonly advertised on TV in Argentina, Costa Rica, and Italy [32, 34, 36]. The commonly advertised foods represent unhealthy foods and beverages that children are often exposed to. Globalization of modern food marketing techniques and the growth of mass media is documented as one of the underlying forces behind the nutrition transition, characterized by increased consumption of energy-dense foods high in sugar and fat such as sweetened beverages and fast foods [37–39]. The extensive exposure to unhealthy foods to children is of concern given the potential links between exposure to unhealthy foods and increased childhood overweight and obesity [40–43]. As such, interventions and programs to curb the increasing trends in overweight and obesity among children and adolescents in Kenya and SSA should incorporate strategies to reduce children's exposure to unhealthy foods through advertisements.

Our qualitative narratives show that children and adolescents are highly attentive to food advertisements. They demonstrated this by providing detailed descriptions of the advertised foods, including the type of food, brand, timing, and the specific TV program during which the advertisement aired. Food advertisements appear to have an immediate impact on brand awareness, recognition, and recall. Over time, this exposure fosters a positive attitude, preference, and normalization of the frequently advertised foods [9]. Consequently, this influences the children's purchase intentions or increases their"pester power"directed at parents. This can lead to the actual purchase and consumption of advertised unhealthy foods by the child/adolescent or their parents [9]. Interestingly, a majority of parents in our study generally perceived the content of food advertisements as safe, provided they did not feature cigarettes or alcohol, products whose advertising and sale are restricted in Kenya. They did not express concern regarding their children's exposure to unhealthy foods. This lack of concern may stem from a gap in knowledge about unhealthy foods and their potential long-term negative health impacts, or a lack of awareness regarding the effects of unhealthy food advertising on children's eating habits. This situation underscores the urgent need for enhanced public education and awareness campaigns focusing on unhealthy foods, including UPFs, and their detrimental effects on children's health. Such initiatives would complement regulatory measures aimed at restricting the marketing and advertising of unhealthy foods. Furthermore, incorporating additional information such as age restrictions and health warnings in unhealthy food advertisements could improve consumer awareness, similar to the impact of warnings on alcohol advertisements noted by participants in this study. Other studies have shown that warning labels on unhealthy foods may deter the selection and purchase of unhealthy foods [44, 45]

Our findings highlight significant use of persuasive techniques in advertisements that target children. Our observations align with findings from several countries, including South Africa, India [46], Colombia, Brazil, and Malaysia, where similar child-focused persuasive strategies are common in food advertising [47]. Existing literature has consistently indicated that children are particularly susceptible to the persuasive impact of unhealthy food advertisements. This vulnerability stems from their less developed cognitive abilities for critically evaluating and understanding the persuasive intent behind these messages, in comparison to adults [48]. Studies have further demonstrated that employing such persuasive techniques can lead to increased preferences for, purchase requests for, and food choices favoring the advertised products among children [20, 49]. This is particularly concerning as these techniques are frequently used to promote unhealthy food options in the advertisements we examined.

The widespread exposure of children to unhealthy foods through advertising is a growing global concern. This is driven by the high rates of unhealthy food consumption among children and the rising trends in overweight/obesity and diet-related non-communicable diseases within this demographic [19, 50]. For instance, in Kenya, adolescent girls exhibit the highest consumption of unhealthy foods like deep-fried and sweet/salty snacks when compared to other women of reproductive age [3].

Consequently, various policy recommendations have been proposed internationally to limit children's exposure to unhealthy food advertisements [19, 51, 52]. While the implementation of such policies is more prevalent in high-income countries, there is limited evidence of their adoption in low- and middle-income countries (LMICs) [53]. However, evidence from countries that have implemented these regulations suggests their potential effectiveness in reducing children's exposure to unhealthy foods [42, 52]. For example, in Chile, restrictions on unhealthy food advertisements on television led to an approximate halving of children's exposure to such products [54]. Notably, voluntary bans and restrictions appear to be less effective in curbing child-targeted unhealthy food advertising compared to mandatory regulations [42]. A review by Kelly et al. (2010) revealed a higher frequency of advertising of unhealthy foods during peak viewing times in countries with industry self-regulated programs compared to countries that have no regulations [10]. A 2016 review of food environment policies in Kenya revealed existing regulations on the marketing of breastmilk substitutes, but a lack of restrictions on marketing unhealthy foods to children and adolescents [55]. Therefore, there is a pressing need to establish a policy or regulations in Kenya that limit children's exposure to unhealthy foods through television advertisements. This would serve as a crucial public health strategy to address the increasing trends in overweight and obesity within the country.

Strengths and limitations

One major strength of this study is that it provides a robust assessment of children's exposure to food advertisements in Kenya, leveraging nationally representative data on TV assessments of both children's and family/general programming that children may watch. Secondly, our 18-h recording window (06:00–00:00) captured a diverse range of TV viewership, encompassing both peak and non-peak hours for children. Thirdly, the inter-rater reliability test implemented during the TV advertisement coding and reporting processes was crucial in minimizing potential bias. Finally, the inclusion of perspectives from children, adolescents, and parents across major cities, secondary cities, and rural areas has significantly enriched our understanding of the persuasive techniques used in food advertising, the impact of these advertisements on children's food behaviors, and the limited awareness among parents regarding prevalent unhealthy food advertising.

We acknowledge that a limitation of our study is the absence of nutritional content information for the advertised foods. Consequently, we were unable to classify the healthiness of these foods based on nutrient profile model thresholds. However, the application of the NOVA classification system allowed us to categorize foods according to their level of processing, with UPF representing the least healthy category. Additionally, COVID-19 restrictions at the time, which limited group meetings to a maximum of 1 h, prevented us from collecting extensive socio-demographic data for participants. Instead, we prioritized in-depth qualitative discussions focused on the study's research questions. We have, however, documented the number of interviews and group discussions conducted at each study school/site, along with the average number of participants per group.

Conclusions

Our study has revealed that the vast majority of food advertisements broadcast on Kenyan television during children's peak viewing hours promote unhealthy foods. Notably, children demonstrate strong engagement with these advertisements, exhibiting clear recall of advertised food brands and the marketing strategies employed.

Food advertisements often utilize various persuasive techniques that effectively capture children's attention. Interestingly, parents generally express limited concern regarding the advertising of unhealthy foods, often perceiving them as safe unless they promote alcohol or tobacco products. This lack of concern may stem from a limited understanding of the risks associated with unhealthy foods or the widespread social acceptance and normalization of such products.

To address the increasing rates of adolescent overweight and obesity in Kenya, there is a clear necessity to develop and implement policies that restrict children's exposure to the advertising of unhealthy foods. It is crucial that these policy interventions are accompanied by public education and awareness campaigns to improve understanding of unhealthy foods and their potential negative impacts on long-term health.

Abbreviations

CAK	Communications Authority of Kenya
ECHO	Ending Childhood Obesity
FGD	Focus group discussions
IDI	In-depth interviews
IDRC	International Development Research Centre
INFORMAS	International Network for Food and Obesity/NCDs Research, Monitoring and Action Support
LMIC	Low- and middle-income countries
NACOSTI	National Commission for Science, Technology and Innovation
NCDs	Non-communicable diseases
SSA	Sub-Saharan Africa
TV	Television
UPFs	Ultra-processed foods
WHA	World Health Assembly

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12916-025-04327-0.

Additional file 1. Key informant interview and focus group discussions guides.

Acknowledgements
The authors acknowledge officers in the Ministries of Health, Ministry of Education in Kenya, and the county governments of Nairobi, Mombasa, and Baringo for supporting the research in the country and in specific counties.

Author Contribution
GA, SV, MH, CA and AL designed study and advised the analysis plans; MNW, CK, VO, and SM Coordinated the project implementation, monitoring, data collection and management; CK and MNW led the analysis and co-drafted the manuscript, which was reviewed and edited by GA, KK-G, CA, EKM, SV, AL,VO, SM and MH. All the authors approved the final version of the manuscript.

Funding
This research is funded by the International Development Research Centre (IDRC) Canada, Grant number: 1019132–001. MNW is supported by a UMC Utrecht Global Health PhD support fellowship. The funders played no role in the conceptualization, design, data collection, analysis, decision to publish, or preparation of the manuscript.

Data availability
The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics approval and consent to participate
This study was approved by the African Medical and Research Foundation (AMREF) ethics and scientific review committee (approval number: ESRC P901/2020) and National Commission for Science, Technology and Innovation (NACOSTI) (license number: NACOSTI/P/21/8429). Ministry of Education approvals/permits were obtained at national, county, sub-county, and school levels for engagement with the students and school community (Nairobi, Mombasa, and Baringo). Written consent was obtained from the adult participants; parental consent and assents from the children/adolescents were obtained before commencement of the interviews.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

Author details
[1]Nutrition and Food Systems Unit, African Population and Health Research Center, Nairobi, Kenya. [2]Julius Global Health, Department of Global Public Health and Bioethics, Julius Center for Health Sciences and Primary Care, University Medical Center Utrecht, Utrecht University, Utrecht, Netherlands. [3]Chronic Disease Management Unit, African Population and Health Research Center, Nairobi, Kenya. [4]Department of Public and Occupational Health, Amsterdam Public Health, University of Amsterdam Medical Center, Amsterdam, the Netherlands. [5]Division of Endocrinology, Diabetes, and Metabolism, Department of Medicine, Johns Hopkins University School of Medicine, Baltimore, MD, USA. [6]UMR MoISA (Montpellier Interdisciplinary Centre On Sustainable Agri-Food Systems), (Univ Montpellier, CIRAD, CIHEAM-IAMM, INRAE, Institut Agro, IRD), Montpellier, France. [7]Department of Population, Family and Reproductive Health, School of Public Health, University of Ghana, P.O. Box LG 13, Legon, Accra, Ghana. [8]Division of Epidemiology and Biostatistics, School of Public Health, Faculty of Health Sciences, University of the Witwatersrand, Johannesburg, South Africa. [9]Service of Lifestyle and Chronic Diseases, Brussels, Belgium. [10]Department of Women's and Children's Health, Karolinska Institutet, Stockholm, Sweden. [11]Department of Global and Public Health, University of Nairobi, Nairobi, Kenya. [12]School of Public Health, University of Witwatersrand, Johannesburg, South Africa.

Received: 30 October 2024 Accepted: 6 August 2025
Published online: 30 October 2025

References
1. World Health Organization. Obesity and overweight - WHO. 2024.
2. Phelps NH, Singleton RK, Zhou B, Heap RA, Mishra A, Bennett JE, et al. Worldwide trends in underweight and obesity from 1990 to 2022: a pooled analysis of 3663 population-representative studies with 222 million children, adolescents, and adults. The Lancet. 2024. https://doi.org/10.1016/S0140-6736(23)02750-2.
3. KNBS and ICF. Kenya Demographic and Health Survey 2022. Demographic and Health Survey 2022. 2023.
4. Bhutta ZA, Norris SA, Roberts M, Singhal A. The global challenge of childhood obesity and its consequences: what can be done? Lancet Glob Health. 2023. https://doi.org/10.1016/S2214-109X(23)00284-X.
5. Shaban Mohamed MA, AbouKhatwa MM, Saifullah AA, Hareez Syahmi M, Mosaad M, Elrggal ME, et al. Risk factors, clinical consequences, prevention, and treatment of childhood obesity. Children (Basel). 2022. https://doi.org/10.3390/children9121975.
6. Choukem S-P, Tochie JN, Sibetcheu AT, Nansseu JR, Hamilton-Shield JP. Overweight/obesity and associated cardiovascular risk factors in sub-Saharan African children and adolescents: a scoping review. Int J Pediatr Endocrinol. 2020. https://doi.org/10.1186/s13633-020-0076-7.
7. Lane MM, Gamage E, Du S, Ashtree DN, McGuinness AJ, Gauci S, et al. Ultra-processed food exposure and adverse health outcomes: umbrella review of epidemiological meta-analyses. BMJ. 2024. https://doi.org/10.1136/bmj-2023-077310.
8. Patton GC, Neufeld LM, Dogra S, Frongillo EA, Hargreaves D, He S, et al. Nourishing our future: the Lancet series on adolescent nutrition. Lancet. 2022. https://doi.org/10.1016/S0140-6736(21)02140-1.
9. Kelly B, King MPsy L, Chapman Mnd K, Boyland E, Bauman AE, Baur LA. A hierarchy of unhealthy food promotion effects: identifying methodological approaches and knowledge gaps. Am J Public Health. 2015;105:e86-95. https://doi.org/10.2105/AJPH.2014.302476.
10. Kelly B, Vandevijvere S, Ng SH, Adams J, Allemandi L, Bahena-Espina L, et al. Global benchmarking of children's exposure to television advertising of unhealthy foods and beverages across 22 countries. Obes Rev. 2019. https://doi.org/10.1111/obr.12840.
11. Sadeghirad B, Duhaney T, Motaghipisheh S, Campbell NRC, Johnston BC. Influence of unhealthy food and beverage marketing on children's dietary intake and preference: a systematic review and meta-analysis of randomized trials. Obes Rev. 2016. https://doi.org/10.1111/obr.12445.
12. Boyland E, McGale L, Maden M, Hounsome J, Boland A, Angus K, et al. Association of food and nonalcoholic beverage marketing with children and adolescents' eating behaviors and health: a systematic review and meta-analysis. JAMA Pediatr. 2022. https://doi.org/10.1001/jamapediatrics.2022.1037.
13. Story M, French S. Food advertising and marketing directed at children and adolescents in the US. Int J Behav Nutr Phys Act. 2004. https://doi.org/10.1186/1479-5868-1-3.

14. WHO. Marketing of Non-Alcoholic Beverages To children : report of a WHO forum and technical meeting, Oslo, Norway, 2–5 May 2006. 2006.
15. GeoPoll. Kenya Media in 2023 – Top TV and Radio Shows. Media Audience Measurement. 2023.
16. Communications Authority of Kenya. Broadcasting Report Baseline Survey for Broadcasting Services in Kenya. 2022.
17. Kigaru DMD, Loechl C, Moleah T, Macharia-Mutie CW, Ndungu ZW. Nutrition knowledge, attitude and practices among urban primary school children in Nairobi City, Kenya: a KAP study. BMC Nutr. 2015. https://doi.org/10.1186/s40795-015-0040-8.
18. World Health Assembly (WHA). Resolution WHA63.14. Marketing of food and non-alcoholic beverages to children. Geneva, Switzerland; 2010.
19. World Health Organization. Report of the Commission on Ending Childhood Obesity: implementation plan. World Health Organization. 2017.
20. Karanja A, Ickowitz A, Stadlmayr B, McMullin S. Understanding drivers of food choice in low-and middle-income countries: a systematic mapping study. Glob Food Secur. 2022;32: 100615.
21. Kelly B, King L, Baur L, Rayner M, Lobstein T, Monteiro C, et al. Monitoring food and non-alcoholic beverage promotions to children. Obes Rev. 2013(S1). https://doi.org/10.1111/obr.12076.
22. Geopoll. Kenya's Television Landscape Throughout Q1, Q2, and Q3 of 2020. Media Audience Measurement. 2020.
23. KARF. Audience Measurement and Industry Trends Report for Q2 2019–2020. The Kenya Audence Research Foundation. 2020;65.
24. Monteiro CA, Cannon G, Levy RB, Moubarac JC, Louzada MLC, Rauber F, et al. Ultra-processed foods: what they are and how to identify them. Public Health Nutr. 2019. https://doi.org/10.1017/S1368980018003762.
25. StataCorp LLC. Stata : Statistical software for data science.
26. Kenya National Beureau of Statistics. KENYA - Kenya Integrated Household Budget Survey 2015–2016. Kenya National Data Archive (KeNADA. 2016.
27. Ersson M, Köbin U. Children ' s Perception of TV-advertising A Case Study of 3rd Graders in Sweden. Sweden: Lulea University of Technology; 2006.
28. Dilogini BK, Shivany S, A DK, Shivany σ S. Influences of Television Advertisement on Children's Behavior: An Experences Base Study. Int Res J Publisher: Global J Inc. 2015;15.
29. Braun V, Clarke V. Qualitative Research in Psychology Using thematic analysis in psychology Using thematic analysis in psychology. Qual Res Psychol. 2006.
30. Yamoah DA, De Man J, Onagbiye SO, Mchiza ZJ. Exposure of children to unhealthy food and beverage advertisements in South Africa. Int J Environ Res Public Health. 2021. https://doi.org/10.3390/ijerph18083856.
31. Gallus S, Borroni E, Stival C, Kaur S, Davoli S, Lugo A, et al. Food advertising during children's television programmes in Italy. Public Health Nutr. 2021;24:4663–70. https://doi.org/10.1017/S1368980020004693.
32. Rovirosa A, Zapata ME, Gómez P, Gotthelf S, Ferrante D. Food and beverage advertising on children's TV channels in Argentina: frequency, duration, and nutritional quality. Arch Argent Pediatr. 2017;115:28–34. https://doi.org/10.5546/aap.2017.eng.28.
33. Speers SE, Harris JL, Schwartz MB. Child and adolescent exposure to food and beverage brand appearances during prime-time television programming. Am J Prev Med. 2011;41:291–6. https://doi.org/10.1016/j.amepre.2011.04.018.
34. Kontsevaya AV, Imaeva AE, Balanova YA, Kapustina AV, Breda J, Jewell JM, et al. The extent and nature of television food advertising to children and adolescents in the Russian Federation. Public Health Nutr. 2020;23:1868–76. https://doi.org/10.1017/S1368980020000191.
35. Matthews AE. Children and obesity: a pan-European project examining the role of food marketing. Eur J Public Health. 2008;18:7–11. https://doi.org/10.1093/eurpub/ckm015.
36. Zamora-Corrales I, Jensen ML, Vandevijvere S, Ramírez-Zea M, Kroker-Lobos MF. Television food and beverage marketing to children in Costa Rica: current state and policy implications. Public Health Nutr. 2019;22:2509–20. https://doi.org/10.1017/S1368980019000776.
37. Baker P, Machado P, Santos T, Sievert K, Backholer K, Hadjikakou M, et al. Ultra-processed foods and the nutrition transition: Global, regional and national trends, food systems transformations and political economy drivers. Obes Rev. 2020. https://doi.org/10.1111/obr.13126.
38. Popkin BM, Adair LS, Ng SW. Global nutrition transition and the pandemic of obesity in developing countries. Nutr Rev. 2012;70:3–21. https://doi.org/10.1111/j.1753-4887.2011.00456.x.
39. Popkin BM. Global nutrition dynamics: the world is shifting rapidly toward a diet linked with noncommunicable diseases. Am J Clin Nutr. 2006. https://doi.org/10.1093/ajcn/84.2.289.
40. Powell LM, Wada R, Khan T, Emery SL. Viewpoint: Food and beverage television advertising exposure and youth consumption, body mass index and adiposity outcomes. Can J Econ. 2017. https://doi.org/10.1111/caje.12261.
41. Andreyeva T, Kelly IR, Harris JL. Exposure to food advertising on television: associations with children's fast food and soft drink consumption and obesity. Econ Hum Biol. 2011. https://doi.org/10.1016/j.ehb.2011.02.004.
42. Coleman PC, Hanson P, van Rens T, Oyebode O. A rapid review of the evidence for children's TV and online advertisement restrictions to fight obesity. Prev Med Rep. 2022. https://doi.org/10.1016/j.pmedr.2022.101717.
43. Norman J, Kelly B, Boyland E, McMahon AT. The impact of marketing and advertising on food behaviours: evaluating the evidence for a causal relationship. Curr Nutr Rep. 2016. https://doi.org/10.1007/s13668-016-0166-6.
44. Bopape M, Taillie LS, Swart R. Perceived effect of warning label on parental food purchasing and drivers of food selection among South African parents–An exploratory study. Front Public Health. 2022;10. https://doi.org/10.3389/fpubh.2022.939937.
45. Clarke N, Pechey E, Kosīte D, König LM, Mantzari E, Blackwell AKM, et al. Impact of health warning labels on selection and consumption of food and alcohol products: systematic review with meta-analysis. Health Psychol Rev. 2020. https://doi.org/10.1080/17437199.2020.1780147.
46. Hebden L, King L, Kelly B. Art of persuasion: an analysis of techniques used to market foods to children. J Paediatr Child Health. 2011. https://doi.org/10.1111/j.1440-1754.2011.02025.x.
47. Ng SH, Kelly B, Se CH, Chinna K, Sameeha MJ, Krishnasamy S, et al. Obesogenic television food advertising to children in Malaysia: Sociocultural variations. Glob Health Action. 2015. https://doi.org/10.3402/gha.v7.25169.
48. Nairn A, Fine C. Who's messing with my mind? The implications of dual-process models for the ethics of advertising to children. Int J Advert. 2008. https://doi.org/10.2501/S0265048708080062.
49. Dos Santos SL, De Rose JCC. Investigating the impact of stimulus equivalence on children's food choice and preference. Temas Psicol. 2018. https://doi.org/10.9788/TP2018.1-01.
50. Khandpur N, Neri DA, Monteiro C, Mazur A, Frelut ML, Boyland E, et al. Ultra-processed food consumption among the paediatric population: an overview and call to action from the European Childhood Obesity Group. Ann Nutr Metab. 2020. https://doi.org/10.1159/000507840.
51. World Health Organization. "Best buys" and other recommended interventions for the prevention and control of noncommunicable diseases. World Health Organisation. 2017.
52. Boyland E, McGale L, Maden M, Hounsome J, Boland A, Jones A. Systematic review of the effect of policies to restrict the marketing of foods and non-alcoholic beverages to which children are exposed. Obes Rev. 2022. https://doi.org/10.1111/obr.13447.
53. Kelly B, Backholer K, Boyland E, Kent MP, Bragg MA, Karupaiah T, et al. Contemporary approaches for monitoring food marketing to children to progress policy actions. Curr Nutr Rep. 2023;12:14–25. https://doi.org/10.1007/s13668-023-00450-7.
54. Correa T, Reyes M, Taillie LS, Corvalán C, Dillman Carpentier FR. Food advertising on television before and after a national unhealthy food marketing regulation in Chile, 2016–2017. Am J Public Health. 2020. https://doi.org/10.2105/AJPH.2020.305658.
55. Asiki G, Wanjohi MN, Barnes A, Bash K, Muthuri S, Amugsi D, et al. Benchmarking food environment policies for the prevention of diet-related non-communicable diseases in Kenya: national expert panel's assessment and priority recommendations. PLoS One. 2020;15: e0236699. https://doi.org/10.1371/journal.pone.0236699.

Publisher's Note